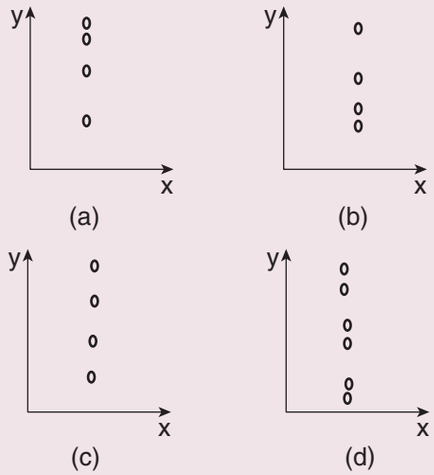
(a) 1 (b) 2

(c) 4 (d) 0.5

10. A ball is dropped from some height towards the ground. Which one of the following represents the correct motion of the ball?



11. If a particle executes uniform circular motion in the xy plane in clock wise direction, then the angular velocity is in

(a) +y direction

(b) +z direction

(c) -z direction

(d) -x direction

12. If a particle executes uniform circular motion, choose the correct statement (NEET 2016)

(a) The velocity and speed are constant.

(b) The acceleration and speed are constant.

(c) The velocity and acceleration are constant.

(d) The speed and magnitude of acceleration are constant.

II. Short Answer Questions

1. Explain what is meant by Cartesian coordinate system?

2. Define a vector. Give examples

13. If an object is thrown vertically up with the initial speed u from the ground, then the time taken by the object to return back to ground is

(a) $\frac{u^2}{2g}$ (b) $\frac{u^2}{g}$

(c) $\frac{u}{2g}$ (d) $\frac{2u}{g}$

14. Two objects are projected at angles 30° and 60° respectively with respect to the horizontal direction. The range of two objects are denoted as R_{30^0} and R_{60^0}. Choose the correct relation from the following

(a) $R_{30^0} = R_{60^0}$

(b) $R_{30^0} = 4R_{60^0}$

(c) $R_{30^0} = \frac{R_{60^0}}{2}$

(d) $R_{30^0} = 2\, R_{60^0}$

15. An object is dropped in an unknown planet from height 50 m, it reaches the ground in 2 s. The acceleration due to gravity in this unknown planet is

(a) g = 20 m s^{-2} (b) g = 25 m s^{-2}

(c) g = 15 m s^{-2} (d) g = 30 m s^{-2}

Answers

1) d	**2)** d	**3)** c	**4)** c	**5)** a
6) a	**7)** b	**8)** c	**9)** a	**10)** a
11) c	**12)** d	**13)** d	**14)** a	**15)** b

3. Define a scalar. Give examples

4. Write a short note on the scalar product between two vectors.

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